

Mids prac2 enter command 1 by 1 and clean this line to remove error

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score, classification_report
```

```
# Load dataset
```

```
data = pd.read_csv(r'D:/MIDS_PRAC/tweets.csv')
```

```
# Clean column names
```

```
df = pd.DataFrame(data)
```

```
df.columns = df.columns.str.strip().str.lower()
```

```
# Features and target
```

```
X = df['text']
```

```
y = df['airline_sentiment']
```

```
# Train-test split
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# TF-IDF vectorization
```

```
vectorizer = TfidfVectorizer()
```

```
X_train_tfidf = vectorizer.fit_transform(X_train)
```

```
X_test_tfidf = vectorizer.transform(X_test)
```

```
# KNN Classifier
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train_tfidf, y_train)
```

```
# Prediction
```

```
y_pred = knn.predict(X_test_tfidf)
```

```
# Evaluation
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nClassification Report:\n", classification_report(y_test, y_pred, zero_division=0))
```